Data Quality Report — ZW-BioBank v1.0

Annex D template · generated from the validator output, not written by hand

This report follows the Annex D template row for row. It is generated by **scripts/generate_reports.py** from **processed/qc_log_v1.csv**, which is itself written by **scripts/validate.py**. Nothing in the table below is typed by hand, so it cannot describe a validation run that did not happen.

Report item	Response
Dataset name	ZW-BioBank v1.0 — Zimbabwe Multi-Modal Indigenous Biology Dataset
Dataset version	v1.0.0 (Phase I specification; records in processed/ are illustrative)
Collection period	Declared window 2026-09-01 to 2027-02-28. Dry season Sep-Nov 2026; wet season refresh Jan-Mar 2027. The validator runs in 'specification' mode, in which a collection date inside the planned window but ahead of today is expected and reported as informational. It becomes a hard failure in production mode.
Number of records or files	120 environmental samples, 55 genomic sequences, 90 metabolomic features, 52 heritage records. 888 quality-control events. Target at completion: 500 samples, 2,500 metabolomic features, 300-500 heritage records.
Completeness summary	Critical-field completeness threshold 95%, applied per table to the field lists in validation/validation_rules.json. All tables pass. 46 field-level completeness checks passed, 0 failed.
Accuracy audit method	Three layers. Automated range and quality checks (preservation latency against a 120-minute target; mean Phred score against Q10; mass error against 5 ppm). A 10% human audit sample against field notebooks and instrument logs. A second bilingual reviewer on every heritage record before release. Duplicate extraction on 10% of samples measures instrument and protocol reproducibility directly.
Consistency checks performed	Identifier format against documented patterns (4 passed). Consent-log agreement on reference, endorsement and withdrawal status (52 passed). Denormalised heritage_link against the parent sample. All categorical fields are constrained to controlled vocabularies in the DDL, so an inconsistent value cannot be written in the first place.
Validity checks performed	Every record is inserted through schema/schema.sql; type, range, controlled vocabulary and cross-field constraints are enforced by the database rather than by script (317 records admitted). Coordinates constrained to Zimbabwe bounds. ISO 8601 on all dates. SHA-256 format on every raw-file reference.
Duplicate records found	None. Primary-key uniqueness confirmed across all tables. Near-duplicate detection on coordinates, date, sample type, depth and collector found no candidate double entries.
Missing critical fields	None below the 95% threshold. Fields left empty are optional and their meaning is documented in the missing_value_rule column of schema/data_dictionary.csv: empty means not applicable or not yet determined, never zero.
Outliers or anomalies	0 sequence record(s) below Q10, flagged provisional rather than discarded. 0 sample(s) above the 120-minute preservation target. Outliers are retained and labelled; discarding them would bias the dataset toward easy collection conditions.
Label quality check	Ten labels defined in labels/label_taxonomy.csv, each with a definition, a positive example and a negative example naming the specific mistake it guards against. Double annotation on every non-unknown compound_class assignment. Second-reviewer sign-off on every novelty call. 18 MSI level 3-4 features carry a warning so they are never reported as identifications. 0 heritage record(s) below human_verified translation status are held out of every release tier.

Report item	Response
Bias and coverage observations	Eight risks registered in governance/bias_risk_register.csv, each with a runnable monitoring query. Ndebele-language heritage records meet the 30% floor. Ecozone coverage is tracked against the 200/150/150 quota, which names three strata and only three. Collection is not confined to them: where a knowledge holder leads the team outside the sampling frame, the sample is recorded with ecozone 'Other' and a free-text locality description rather than being refused or relabelled. Those records are admitted and validated in full but never counted toward the quota, and check V24 reports their share on every run so drift toward convenience collection is visible rather than absorbed. Any ecozone carrying records from a single season is flagged as a warning on every run and clears when the seasonal refresh is collected. Live coverage is queryable through view v_bias_coverage.
Privacy checks	No direct identifiers detected in free-text fields (patterns for telephone and national ID numbers are scanned on every ingest). Every public-tier record confirmed masked to ward centroid with declared precision. Holder identity is absent from the dataset at every tier; it resides only in the AES-256 encrypted Holder Identity Register in SIRDC custody. Consent gate, prior-consent ordering and withdrawal propagation are all enforced and all tested.
Known limitations	Phase I is a 500-sample pilot, not a national inventory. Untargeted LC-MS confidently identifies only a minority of detected features. The heritage corpus reflects consenting holders in participating communities and is not representative of Zimbabwe. Nanopore results are reproducible only against the recorded basecalling model. ICD-11 mappings of traditional indications are approximate navigation aids, not clinical equivalences. Sacred knowledge is absent by design, not by oversight.
Corrective actions	0 failing check(s) at this run. Warnings carry a documented caveat and a named issue owner in validation/error_log.csv; each is tracked to resolution in the qc log, which is retained permanently. A critical finding cannot be waived without a recorded action and owner — enforced by constraint on the qc_log table.
Validation status	PASS — 870 pass, 18 warning, 0 fail across 888 checks and 16 distinct check types. All seven quality dimensions exercised: accuracy, completeness, consistency, integrity, timeliness, uniqueness, validity.

Appendix A · Checks performed

Every check is defined in validation/validation_rules.json with its quality dimension, severity and pass condition. Thresholds are read from that file, so a reviewer can change one and re-run without touching code.

Check	Dimension	Pass	Warning	Fail
annotation_confidence_honesty	accuracy	0	18	0
checksum_format	integrity	145	0	0
collection_window	timeliness	120	0	0
consent_log_agreement	consistency	52	0	0
critical_field_completeness	completeness	46	0	0
ecozone_coverage	completeness	3	0	0
identifier_format	consistency	4	0	0
language_representation	completeness	1	0	0
location_masking	validity	120	0	0
near_duplicate_sample	uniqueness	1	0	0
no_direct_identifiers	validity	52	0	0
out_of_frame_coverage	completeness	1	0	0
primary_key_uniqueness	uniqueness	4	0	0
schema_insert	validity	317	0	0
seasonal_coverage	completeness	3	0	0
source_id_resolves	integrity	1	0	0

Appendix B · Open findings

Result	Severity	Record	Finding	Owner
warning	low	ZW-MET-2026-025	MSI level 4: 'Unknown sesquiterpenoid dimer' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-026	MSI level 4: 'Unknown polyketide' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-027	MSI level 3: 'Triterpenoid saponin (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-028	MSI level 3: 'Anthraquinone glycoside (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-029	MSI level 3: 'Sesquiterpene lactone (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-030	MSI level 4: 'Unannotated feature' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist

Result	Severity	Record	Finding	Owner
warning	low	ZW-MET-2026-055	MSI level 4: 'Unknown sesquiterpenoid dimer' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-056	MSI level 4: 'Unknown polyketide' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-057	MSI level 3: 'Triterpenoid saponin (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-058	MSI level 3: 'Anthraquinone glycoside (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-059	MSI level 3: 'Sesquiterpene lactone (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-060	MSI level 4: 'Unannotated feature' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-085	MSI level 4: 'Unknown sesquiterpenoid dimer' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-086	MSI level 4: 'Unknown polyketide' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-087	MSI level 3: 'Triterpenoid saponin (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-088	MSI level 3: 'Anthraquinone glycoside (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-089	MSI level 3: 'Sesquiterpene lactone (putative)' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist
warning	low	ZW-MET-2026-090	MSI level 4: 'Unannotated feature' is a putative annotation and must not be reported as an identification.	Pharmaceutical Chemist